Appendix A1. Prompt / research log

AI assistance was used for literature discovery, comparison, drafting, figure/table preparation and technical critique. The entries summarize the questions actually addressed during this assisted workflow; they are not a claimed verbatim transcript. Scientific support comes from the listed publications and datasets, not the AI tool.

Assignment prompt	Research question addressed	Recorded output
1. Define the study area	Compare Eel River, Point Arena, onshore Humboldt, assessment province and State/Federal definitions.	Section 2 and Figure 1.
2. Build the source library	Find and audit technical literature with direct agency/publisher identifiers.	References and Appendix A5.
3. Geologic history	Link deposition, subsidence, uplift and triple-junction migration to petroleum processes.	Section 3, Table 2.
4. Stratigraphy	Keep the two basin successions distinct and qualify petroleum roles.	Section 4, Table 3.
5. Source and maturity	Separate organic richness, maturity, fluid origin and demonstrated correlation.	Section 5, Table 4.
6. Reservoir / seal / trap	Keep compiled onshore properties separate from offshore play concepts.	Section 6, Table 5 and Figure 2.
7. Migration and timing	Build a connected source-to-trap argument and identify missing charge dates.	Section 7 and Figure 3.
8. Thermal regime	Inspect original Site 1019 temperatures, depth datum, fitting method and conductivity.	Section 9, Table 7 and Figure 4.
9. Geothermal significance	Test whether heat, transmissivity, fluids and infrastructure support a resource claim.	Section 10, Table 8.
10. Skeptical review	Challenge basin-wide extrapolation, seep interpretation, trap timing and unverified source data.	Section 11 and Appendix A4.
11. Reference audit	Check authors, titles, report identifiers, DOI/official links, and claim-specific locators.	References and Appendix A2-A5.

Evidence and review conventions

Observation: a reported measurement, sample, well result or mapped feature. **Interpretation:** a publication’s geological explanation. **Inference:** a conclusion drawn in this report from the available evidence. Figures 1-3 include schematic interpretation; Figure 4 replots reported temperatures.

Aly Sadek confirmed personal verification of all twelve audited claims and their cited source sections on **October 20, 2026**. The student verification column records this attestation. Assistant inspection and access limitations are recorded separately; abstract-only access and unavailable tables describe assistant access.

AI-use disclosure: research, synthesis, drafting and document production were AI assisted. No fieldwork, laboratory analysis, instructor approval or student source review is fabricated.

Appendix A2. Claim-source verification

Locators use printed publication page numbers, not PDF viewer indices. Student verification: Aly Sadek attested all twelve claims on October 20, 2026. Assistant checks are recorded separately.

Load-bearing claim	Exact source / locator	Evidence and assistant check	Student verified?
1. The offshore assessment pair spans different geological provinces.	[1] sections 7.4.2-7.4.3, pp. 69-71; Figs. 18-19.	Agency interpretation Original report inspected	Yes
2. Onshore Eel River margins include the Little Salmon thrust and Russ fault.	[2] section 3.2.4.1, p. 32.	Mapped framework Original text inspected	Yes
3. Point Arena coastal stratigraphy differs from the Eel River forearc succession.	[3] author abstract; [4] pp. 419-422, text-Figs. 1-2; [2] pp. 32-33.	Stratigraphic interpretation [2] original; [3] abstract; [4] indexed original text	Yes
4. CGS compiles Rio Dell porosity 22-28% and permeability 1-12 mD.	[2] p. 33, Rio Dell paragraph. Underlying DOG/Stanley data are a secondary compilation.	Historical property compilation Original compilation inspected	Yes
5. The four offshore Eel River wells summarized by BOEM provide only limited hydrocarbon evidence.	[1] p. 69; OCS-Petty P 0019 #1 gilsonite-vein paragraph.	Well-result summary Original report inspected	Yes
6. Point Arena has oil shows in three offshore exploratory wells.	[1] p. 70, paragraph after area description.	Well-result summary Original report inspected	Yes

Audit rule

A bibliographic match establishes that a publication exists. It does not establish that a particular claim is supported. Numerical claims above use inspected original report text or tables. Abstract-only sources are retained for limited qualitative context and are explicitly labeled.

No values from paywalled or inaccessible TOC, Rock-Eval, vitrinite-reflectance or gas-isotope tables have been reconstructed. The absence of such values in this report records an evidence limit rather than a zero value or proof that no source potential exists.

Appendix A3. Claim-source verification (continued)

Locators use printed publication page numbers, not PDF viewer indices. Student verification: Aly Sadek attested all twelve claims on October 20, 2026. Assistant checks are recorded separately.

Load-bearing claim	Exact source / locator	Evidence and assistant check	Student verified?
7. Southern Cascadia faults and folds focus fluid flow and seepage.	[8] sections 3.3-3.4 and 4; Figs. 4-6.	Observation plus interpretation Original full text / figure caption inspected	Yes
8. The Eel River gas study allows both thermogenic and microbial origins.	[6] official USGS author abstract. Detailed isotopic table not accessed.	Published synthesis Official abstract only; numerical isotope claims excluded	Yes
9. 1019C downhole values are 5.8, 6.7 and 8.1°C at 36.8, 55.8 and 74.8 mbsf.	[7] p. 370, Heat Flow; Figs. 19-20, p. 376.	Measured records with fitted temperatures Original chapter inspected	Yes
10. 1019C has a reported 57°C/km gradient and measurement-based 57 mW/m² heat flow.	[7] p. 370; mean measured conductivity 0.989 W/(m·K).	Derived from measurements Original chapter and dimensional calculation inspected	Yes
11. Most 1019C TOC values are 0.5-1.5 wt%, with an uppermost value near 2.2 wt%.	[7] p. 369 and Table 14, p. 373; 2.18 wt% at 0.29 mbsf in printed table.	Measured sediment geochemistry Original chapter / printed table inspected	Yes
12. Gorda Escarpment emergence near 3.7-2.5 Ma is a local history, not a basin-wide charge date.	[9] section 5, especially Figs. 10-11 and unconformity discussion.	Seismic/sample interpretation Original full text inspected	Yes

Audit rule

A bibliographic match establishes that a publication exists. It does not establish that a particular claim is supported. Numerical claims above use inspected original report text or tables. Abstract-only sources are retained for limited qualitative context and are explicitly labeled.

No values from paywalled or inaccessible TOC, Rock-Eval, vitrinite-reflectance or gas-isotope tables have been reconstructed. The absence of such values in this report records an evidence limit rather than a zero value or proof that no source potential exists.

Appendix A4. Corrections and reflection

Statement challenged	Correction made after source checking
“Site 1019 has four downhole Adara measurements.”	The final chapter lists three downhole determinations and a bottom-water datum, giving four points for the gradient [7], p. 370. The report and plot use that distinction.
“A high offshore gradient establishes geothermal power potential.”	57°C/km is a shallow local gradient in low-conductivity sediment. The reported heat flux is 57 mW/m², and no deep flow test is supplied [7]. Resource language was narrowed to screening.
“A microbial mat means the seep methane was generated microbially.”	The cited microbiological paper measures methane oxidation [12]. Gas generation and gas consumption are different processes. Origin remains constrained by separate geochemistry [6].
“One Yager age can be copied without checking.”	[2] calls the Yager Formation Cretaceous; [5] explicitly treats a Paleogene Yager complex. The discrepancy is recorded, and no unsupported numerical burial history is built from it.
“The heat-flow calculation exactly reproduces the published result.”	The rounded inputs give 56.4 mW/m², while the chapter reports 57 mW/m². The report preserves the source value and discloses the calculation difference.

These are corrections or qualifications made during the assisted synthesis and audit. They are not presented as independently performed student actions.

Reflection

The assisted workflow was most useful for separating questions that initially appeared to be one basin study. It organized boundary definitions, linked petroleum-system elements to different kinds of evidence, and made it easier to compare productive fields with unsuccessful offshore tests. It also located a basin-specific thermal chapter with actual downhole values, which made a measured temperature-depth figure possible. The evidence tables exposed several places where a fluent explanation could have become stronger than its support. In particular, methane-consuming microbes cannot identify the original generation mechanism of the gas, and a shallow sediment gradient cannot establish deep geothermal output. Checking the original drilling chapter also corrected the measurement count and clarified the depth datum. Human geological judgment remains necessary to evaluate source correlation, seismic continuity, fault behavior and the timing of charge relative to trap growth. Those decisions require examining the original figures and deciding whether a regional interpretation is transferable to a particular prospect. The report therefore keeps interpretations and proposed tests distinct from measurements, retains negative exploration evidence, and avoids calculating a resource from missing reservoir data. The audit records assistant inspection separately from personal student review, so the workflow remains transparent about who performed each check.

Reflection length: 199 words. The assignment requests 150-250 words.

Appendix A5. Source library and access audit

Full citations and DOI/report numbers are in References. This inventory supplies source type, topic, why it matters and access limitations. “Direct” identifies evidence within the focus area; “regional” indicates contextual transfer.

Source / type	Topic and role	Scope / inspection limit
[1] Agency technical report	Assessment boundaries, offshore wells and plays	Direct, both basins. Original report, sections 7.4.2-7.4.3 and Figs. 18-19
[2] Agency technical report	Onshore structure, stratigraphy, reservoir compilation	Direct, onshore Eel River. Original report, section 3.2.4.1 and Figs. 8-9
[3] Peer-reviewed article	Two Neogene sequences and regional basin architecture	Direct, Eel River. Author abstract and metadata on USGS record. Detailed paywalled figures not used
[4] Peer-reviewed article	Point Arena formation names, biostratigraphy and missing section	Direct coastal outcrops, offshore extrapolation limited. Publisher-indexed original text and metadata. Full page images unavailable
[5] Peer-reviewed article	Yager age terminology, gas-prone organic matter and maturity	Direct outcrop study, buried source correlation unproven. Publisher metadata and abstract indexed by OSTI. Numerical tables not used
[6] Agency technical report	Gas occurrences, microbial and thermogenic contributions	Direct local gas sampling. Official USGS abstract/metadata. Original PDF could not be opened; no isotope numbers used
[7] Scientific drilling technical report	Measured temperatures, conductivity, organic carbon and depth datum	Direct, one shallow offshore site. Original chapter, Heat Flow and Organic Geochemistry sections; Figs. 19-20 and Table 14
[8] Peer-reviewed article	Seep structure, imbricated thrusts, faults and fluid pathways	Direct southern Cascadia, broader than Eel River. Original full text, sections 3.3-3.4 and 4; Figs. 4-6
[9] Peer-reviewed article	Seismic stratigraphy, unconformity and uplift/subsidence	Direct Gorda Escarpment / Vizcaino Block, regional to Point Arena. Original full text, sections 4-5; Figs. 7-11
[10] Peer-reviewed article	Regional observed heat-flow comparison and transient thermal modeling	Regional onshore Coast Ranges, not a Point Arena reservoir test. Original full text, paragraphs 6-12; Figs. 1-4
[11] Peer-reviewed article	Modern tomography and 3-D transition from subduction to translation	Regional tectonic framework. Original full text, sections 1.1-1.2 and 3
[12] Peer-reviewed article	Methane consumption, seep geochemistry and biological alteration	Direct Eel River seep samples with other-basin comparison. Original full text, methods and results; Tables 1-2
[13] Agency jurisdiction webpage	State submerged-land jurisdiction	California-wide boundary context. Official agency text

References [1]-[12] satisfy the substantial technical-source inventory. [13] supports jurisdiction only. None is an AI-generated scientific reference. Original figures/maps to inspect: [1] Figs. 18-19; [2] Figs. 8-9; [7] Figs. 19-20; [8] Fig. 6; [9] Figs. 7-11.